

CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title: AI-Assisted Triageing System of Middle-Acuity Patients in a Resource-Constrained Emergency Department: A 12-Week Local Validation and Shadow-Testing Pilot

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Statement of the Problem

(89 words)

The middle acuity band accounts for the majority of patients at Mercer's emergency department. Existing triage protocols cannot reliably distinguish patients in this band who will deteriorate from those who will not. Published AI triage models report high classification accuracy (1,2) but are validated on single, well-resourced hospital datasets and rarely deployed prospectively (3). This project will develop validate a predictive model on Mercer triage data, then evaluate it in shadow mode alongside nurse triage, to assess whether an AI-assisted system can improve identification of middle-band patients requiring up-triage.

Previous Work

A lot of people have worked on this in the last five years. Most of them agree that AI can sort ED patients better than normal triage, mostly in the crowded middle band. But two problems show up repeatedly, and those are the openings for a pilot at Mercer. Below are seven recent papers, then the two gaps they leave.

Summaries of the Literature

1. **Araouchi and Adda (2024), TriageIntelli (1)** . Araouchi and Adda evaluated seven ML classifiers and a stacking ensemble trained to predict KTAS urgency levels, targeting triage overload in resource-limited settings. The stacking model achieved 80.05% accuracy with AUC of 0.93, outperforming all individual classifiers. The authors state that a key limitation is the reliance on a single dataset, which may hinder the generalisability of the models to other clinical environments with varying demographics, seasonal factors, and hospital contexts. For healthcare AI, this provides a strong accuracy benchmark but cannot be applied to a different clinical population without local validation
2. **Phuttharaksa (2025), Bang Khla Hospital, Thailand (2)**. Phuttharaksa evaluated ML classifiers for ED triage at Bang Khla Hospital, a resource-constrained community facility, using 29,389 cleaned records from 2021 to 2024. A Random Forest achieved 98.67% overall accuracy; performance on the urgent acuity level was F1 of 0.86. The author states that data is from a single hospital only, EMR data quality and completeness may affect results, key features including consciousness level and pain score were unavailable, and the model has not been tested in real-world clinical settings. This provides the most directly relevant precedent for a Mercer pilot, but retrospective performance alone does not confirm clinical readiness.

3. **Tyler et al. (2024), scoping review (3).** Tyler et al. conducted a scoping review of AI and ML applications in ED triage, searching EMBASE, Ovid MEDLINE and Web of Science for peer-reviewed primary studies from US journals, 2013 to 2023. Results demonstrated improvements in triage accuracy, disease identification and risk assessment, with reductions in under- and over-triage rates. The authors state that restricting the search to US publications limits generalisability, and that overconfident model outputs carry patient safety implications. This evidence base does not extend to non-Western or resource-constrained ED settings, limiting direct applicability to a Caribbean context.
4. **Da'Costa et al. (2025), narrative review (4).** Da'Costa et al. reviewed AI triage literature from 2015 to 2024 across PubMed, Scopus, IEEE Xplore and Google Scholar, assessing AI's capacity to address ED overcrowding and inconsistent triage. Reported benefits include improved prioritisation, reduced wait times and NLP-based processing of free-text complaints. The authors state that the review's narrative design lacks the rigour of a systematic review, is restricted to English-language publications, and does not address region-specific infrastructure differences. They also identify data quality, algorithmic bias and low clinician trust as primary barriers to adoption.
5. **Taylor et al. (2025), real-world deployment (5).** Taylor et al. deployed an AI triage decision support tool across three EDs, analysing outcomes across 174,648 visits before and after implementation. The system identified that more than 10% of ESI level 3 patients required up-triage, with model AUC of 0.73 to 0.92. Stated limitations include a quality-improvement study design, single health system scope and variable nurse compliance with tool recommendations. This is the only study reviewed that measures live prospective clinical impact, making it the strongest available evidence of real-world applicability.
6. **De Freitas et al. (2020), Caribbean ED patient flow (6).** De Freitas et al. mapped patient flow in an emergency department in Trinidad and Tobago using non-participant observation, process mapping and informal interviews. They characterised the ED as a complex adaptive system and identified organisational factors that delay flow. The study is qualitative and process-focused, and tests no predictive or triage model. It establishes the Caribbean ED context directly relevant to Mercer, but leaves model-based re-stratification of acuity unaddressed.
7. **Liu et al. (2021), prospective ML triage (7).** Liu et al. developed a machine learning triage algorithm to detect patients needing critical care, then tested it in a prospective cohort that alternated standard and model-assisted triage over matched periods. This is one of few studies to evaluate an ML triage tool prospectively at the point of care rather than on retrospective data alone. The work was single-site and conducted in a well-resourced tertiary hospital. It did not target a middle-acuity band or a Caribbean population.

Identified Gaps

Gap 1: Single-site validation means there is no evidence of generalisability across different clinical populations or resource settings.

All five AI studies report results from a single site or health system. Tyler et al. restrict their review to US publications and cite limited generalisability as a stated weakness (5). TriageIntelli is trained and evaluated on one Korean dataset (1). Phuttharaksa's model is built on one Thai hospital's records (2). Taylor's deployment is confined to one US health system (5). De Freitas et al. document patient flow in a Caribbean ED but evaluate no predictive model (6). No study examines model performance on a Caribbean, resource-constrained, mixed-acuity ED. This gap is directly addressable by training and evaluating on Mercer's own data.

Gap 2: Prospective evaluation of AI triage tools in resource-limited settings is absent from the current literature.

Four of the five AI studies are wholly retrospective (1,2,4,5). Taylor et al. and Liu et al. (7) provide the only prospective evidence, but both operate in well-resourced systems with mature electronic health records, in the United States and China respectively. Da'Costa et al. (4) identify clinician trust and workflow integration as primary unresolved challenges in AI triage adoption. These challenges cannot be evaluated from retrospective accuracy metrics alone. No study has conducted a prospective, point-of-care evaluation in a resource-constrained or Caribbean facility.

Proposed Solution

The proposed solution applies the deployment framework of Taylor et al. (3) to the Mercer context across four phases.

Phase 1: Model development. A predictive model will be trained on Mercer's historical triage data using vital signs, chief complaint and patient disposition as inputs (3). Logistic regression and gradient boosting classifiers will be compared, with interpretability prioritised to support clinician trust, consistent with the concerns raised by Da'Costa et al. (4).

Phase 2: Local validation. The model will be evaluated on a held-out Mercer dataset. This directly addresses Gap 1. Poor performance at this stage will be identified before any clinical involvement.

Phase 3: Shadow-mode evaluation. The model will operate alongside live nurse triage for two to four weeks, logging predictions without influencing clinical decisions (3). Predictions will be compared against actual patient outcomes post-hoc. This directly addresses Gap 2 and provides the prospective, point-of-care evidence currently missing from the literature (2,3).

Phase 4: Final deliverable. At week 12, the project will produce a locally validated model, shadow mode performance metrics and a Board-level go or no-go recommendation.

References

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Key Terms

Plain meanings for the terms used in this proposal, for any reader who is not from a machine learning background.

Term	What it means
Triage	Sorting patients by how urgently they need care, so the sickest are seen first.
Acuity	How severe or urgent a patient's condition is. High acuity means very sick or unstable.
Acuity band / level	The category a patient is placed in based on acuity. The middle band is moderate, neither clearly critical nor clearly minor.
ESI (Emergency Severity Index)	A common five-level triage scale used to rank how urgent a patient is.
KTAS (Korean Triage and Acuity Scale)	A similar five-level triage scale used in Korea.
Up-triage	Moving a patient to a higher urgency level than first assigned, because they are sicker than they looked.
Under-triage	Wrongly assigning a sick patient too low an urgency. This is dangerous.
Over-triage	Wrongly assigning a low-risk patient too high an urgency. This wastes resources.
Machine learning (ML)	A type of AI that learns patterns from examples in data rather than from fixed rules.
Model	The trained program that takes patient data and outputs a prediction.
NLP (Natural Language Processing)	AI that reads and understands free text, such as a nurse's written notes.
Clinical Decision Support (CDS)	A tool that gives clinicians a suggestion to help them decide. It does not replace them.
Accuracy	The share of predictions the model got right.
AUC (Area Under the Curve)	A score from 0.5 to 1.0 for how well a model separates two groups. Higher is better. 0.5 is a coin flip.
F1 score	A score from 0 to 1 that balances how often the model is right against how often it misses cases. Higher is better.
Retrospective	A study that uses data already collected in the past.
Prospective	A study that collects data going forward, in real time.

Term	What it means
Shadow mode	Running the tool live but only logging its suggestions, without changing patient care, to see how it would have done.
Chief complaint	The main reason the patient says they came in.
Vital signs	Basic body measurements such as heart rate, blood pressure, temperature and oxygen level.
Disposition	What happens to the patient after the ED, for example admitted, discharged or sent to intensive care.
Stacking / ensemble model	Combining several models so the group predicts better than any single one alone.